

SIMD Performance Analysis

To evaluate the benefits of SIMD vectorization, we compare a scalar implementation with a SIMD implementation using AVX2 intrinsics. Each experiment was repeated 100 times to obtain reliable estimates.

The SIMD implementation adapts its strategy based on the number of features: for smaller vectors, a single AVX2 accumulator is used, while for larger vectors, multiple accumulators are employed to increase instruction-level parallelism. In both cases, dot products are computed using fused multiply-add operations on 256-bit registers, followed by a reduction to obtain the final sum.

Results

Table 1: Runtime comparison between scalar and SIMD implementations.

Dataset	#Features	Scalar Avg	Scalar Std	SIMD Avg	SIMD Std	Speedup
calhouse	8	0.09913	0.00510	0.03575	0.00495	2.77x
cpusmall	12	0.05432	0.00456	0.02240	0.00210	2.43x
diamonds	26	0.72283	0.04711	0.16547	0.01432	4.37x
allstate	130	12.4938	0.71450	4.62288	0.20676	2.70x
mnist_5vall	784	27.6994	1.45640	10.2133	0.87951	2.71x

Ball Possession Analysis

Histogram of Ball Possession

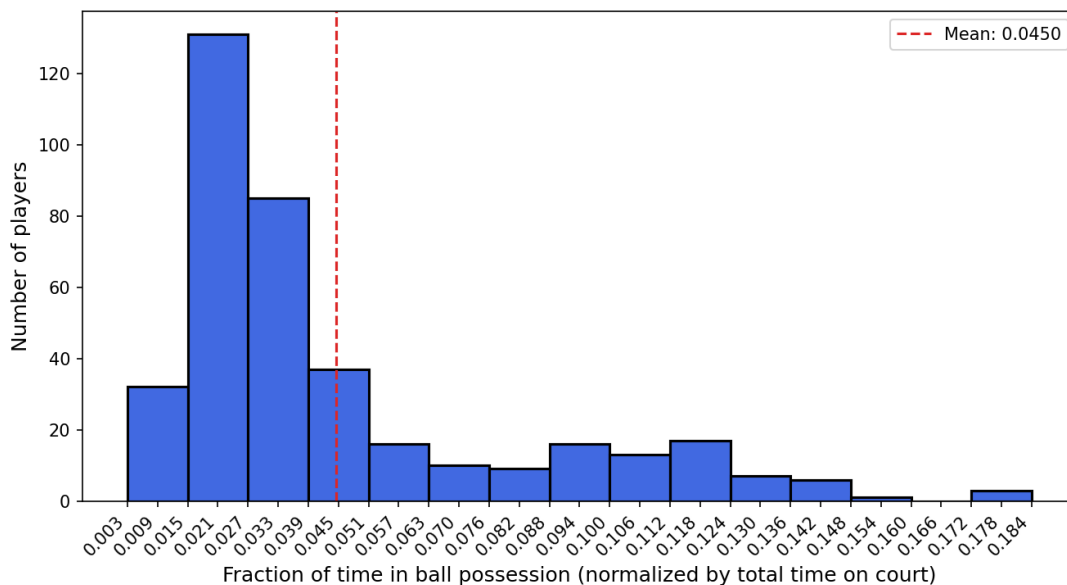


Figure 1: Distribution of normalized ball possession time across players.

Discussion

The histogram illustrates the distribution of ball possession among players. Typically, only a small number of players dominate possession, while most players spend relatively little time in control of the ball.

As with the distance traveled query, in the ball possession query, when calculating distances, large jumps between non-consecutive moments are considered. After doing some EDA on the sample data, it was observed that the majority of consecutive moments do not have gaps larger than 0.5 seconds, and because of that, it is assumed that any two

moments that have a gap larger than 0.5 seconds are most likely a player substitution or a discrepancy in the data. Therefore, distances between moments with large gaps are not included.

Clutch Time Shot Analysis

Stephen Curry Shot Chart

Figure 2 visualizes all field-goal attempts taken by Stephen Curry during clutch time. All shots are transformed so that they are displayed as attempts toward the same basket.

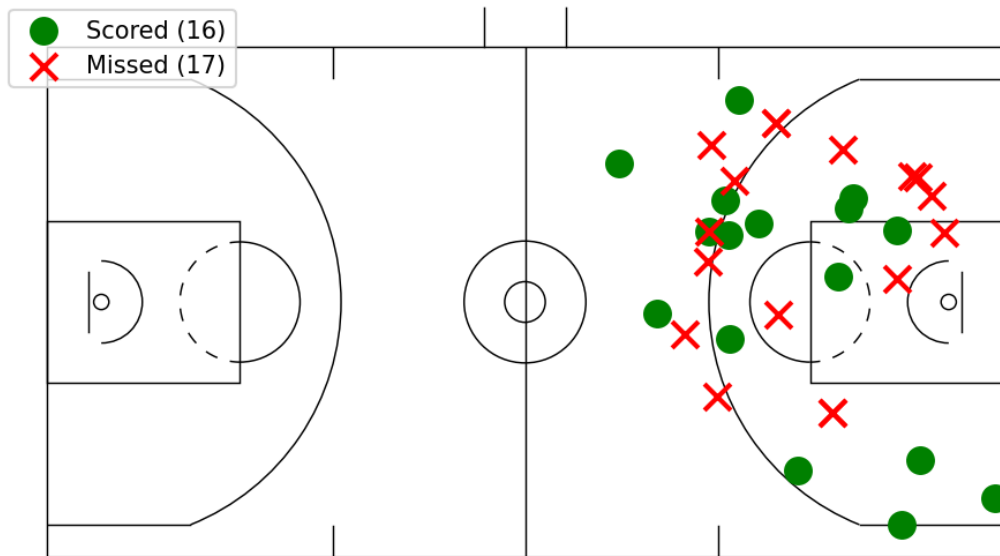


Figure 2: Shot chart for Stephen Curry during clutch time (green = made, red = missed).

Discussion

To determine the location of the clutch shots, a threshold-based approach was used. Because events and moments are not perfectly aligned, it is assumed that the shot location is when the ball hits the 2m mark or its highest point during the event. This is based on most NBA players being above 1.9m, so when winding up, they raise their hands and jump, making the 2m mark approximate the shot location. If the ball never hits 2m, the fallback is to use the event's highest ball point, assuming the shooter was blocked or stripped while winding up.

Execution Details

The `.sh` scripts for the Spark code are intended to be executed from the Spark directory on the cluster. The output is written to `/user/r1035493/output` on the distributed file system.

Each script can be run independently. For first-time execution, the scripts include the `mvn clean package` command to ensure the project is rebuilt before execution.